
PROJECT SUBMISSION DOCUMENT

VayuNetra

AI-Powered Urban Air Quality Intelligence for Smart City Intervention

India already measures its air — 900+ CAAQMS stations — yet a 2024 CAG audit found only 31% of monitored cities have any actionable response protocol. The data exists; the intelligence layer to act on it does not. VayuNetra is that missing **operate** layer: a six-agent AI platform that traces PM2.5 to its sources square-kilometre by square-kilometre, forecasts 72 hours ahead, generates cited enforcement action for officers, and delivers health advisories in four languages to the people breathing it — live in three cities, on ₹0 infrastructure.

Omkar Kadam · Sejal Kumbhar · Abhinav Prasad

Full-Stack AI Engineers

Event ET AI Hackathon 2026 · Problem Statement 5

Theme Smart Cities · Environmental Intelligence · Geospatial Analytics · Public Health

Cities live Delhi · Bengaluru · Mumbai · **Date** 21 July 2026

SECTION 1

Submission links

Live application (frontend)	https://vayunetra-aqi.vercel.app Deployed on Vercel. Landing page → “Open console” → the full operations console.
Backend API	https://vayunetra-c8i8.onrender.com FastAPI on Render. Health check at <code>/health</code> ; 31 HTTP routes + 1 WebSocket.
Source code repository	https://github.com/omkarrr88/VayuNetra Full source, migrations, tests, validation notebook and documentation.
Demo video	https://drive.google.com/file/d/1y5FUBnXGn4iHEeUjkV7_OYeh3nOmNNTK/view?usp=drivesdk
Pitch deck (PDF)	https://drive.google.com/file/d/1huTSRp7aQjCSWy1ueHqZKMoVvgl2S7aB/view?usp=drivesdk
Citizen Telegram bot	@aqivayu_bot — send <code>/start</code> , choose a city, receive live advisories

Reviewer quick path (60 seconds): open the live app → click **Open console** → click any coloured hexagon on the map (opens its *Cell Story*: who is to blame, the model’s evidence, and the 72-hour outlook) → open the **Enforcement** section → click **Evidence dossier** on the top card (real Sentinel-2 satellite patch + regulatory citations) → click **Notice PDF** (a draft enforcement notice with a projected-impact chart) → open **Advisories**, switch the language to Hindi, and click the **IVR call** tab.

SECTION 2

The problem

India’s air quality crisis is a national urban crisis, not a Delhi problem. In 2024–25 Delhi averaged an AQI of 218 (‘Poor’ or worse on 200+ days); Mumbai recorded dangerous AQI on 60+ days in 2024; Bengaluru and Chennai — long considered clean — are measurably deteriorating as vehicle density and construction surge. CPCB data shows **24 of India’s 50 most polluted cities are Tier-1/Tier-2 urban centres**, and *The Lancet Planetary Health* attributes **1.67 million premature deaths a year** to air pollution in India.

Despite 900+ CAAQMS stations deployed under the National Clean Air Programme, a 2024 CAG audit found **only 31% of cities with monitoring data had any actionable multi-agency response protocol** linked to those readings. City administrations do not need another dashboard. They need three things that do not exist together today:

- **Geospatial attribution** — which sources are responsible at this location, right now.
- **Predictive forecasting** — what AQI will be in 24–72 hours at ward/1 km resolution.
- **Enforcement intelligence** — where to send inspectors for maximum impact, with evidence.

SECTION 3

The solution — one closed loop, four stages

VayuNetra fuses monitoring-station data, satellite imagery, mobility feeds, meteorological forecasts and geospatial land-use layers into a single operational loop. The spatial unit everywhere is an **Uber H3 resolution-8 cell (~1 km²)**, so every layer — measurement, blame, forecast, enforcement, advisory — is addressable at the same hyperlocal grain.

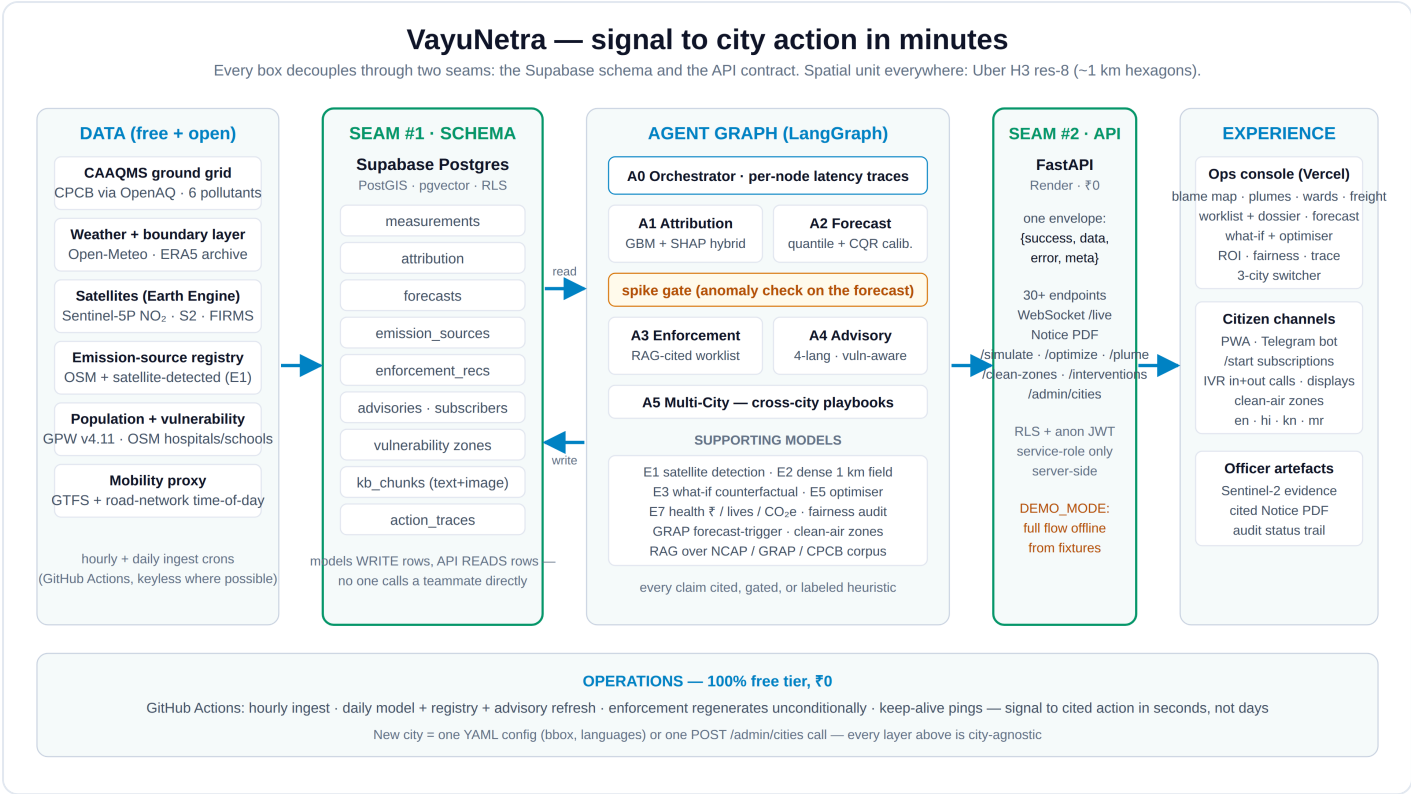
Stage	What the platform does
1 • Trace	Attributes PM2.5 in each ~1 km ² cell to source categories (traffic, construction dust, industrial, biomass burning, transported, other) using a gradient-boosting model with SHAP explanations and statistical confidence scores. Where the model lacks out-of-sample skill, it abstains and falls back to cited chemical-signature priors rather than over-claiming.
2 • Predict	Produces 24/48/72-hour PM2.5 forecasts per cell with conformalised (CQR-calibrated) 80% uncertainty bands, always shown against a persistence baseline. Forecasts drive intervention scheduling, including forecast-triggered GRAP staging.
3 • Act	Correlates hotspots with registered and satellite-detected emission sources and generates a ranked, evidence-backed enforcement worklist. Each recommendation carries an evidence dossier (real Sentinel-2 satellite patch + RAG-retrieved regulatory citations) and a one-click draft Notice PDF. Officer-in-the-loop throughout; dispatching arms automatic before/after impact measurement.
4 • Protect	Generates ward-level citizen health advisories in English, Hindi, Kannada and Marathi, targeted using 2,624 vulnerability-scored zones (hospitals, schools, elder-care, outdoor-work sites × real population), and pushes them through the web app, a live Telegram bot, real IVR phone calls and public-display boards.

Measured end-to-end latency from pollution signal to cited action: 0.8–9.7 seconds, traced per agent node in-product, against a north-star of five minutes and a status quo of days.

Production scale, running live

Metric	Live value	Metric	Live value
Cities operating	3 (Delhi, Bengaluru, Mumbai)	Modeled ~1 km ² cells	6,394
Emission sources tracked	547 registered + satellite-detected	Vulnerability-scored zones	2,624
Enforcement recommendations	390, all RAG-cited	Advisory languages	4 (en · hi · kn · mr)
API surface	31 HTTP routes + 1 WebSocket	Automated tests	169 backend + 6 end-to-end
Infrastructure cost	₹0 (free tiers end-to-end)	Satellite evidence coverage	390 / 390 dossier sources

Architecture



VayuNetra system architecture — data sources → Supabase schema (**seam #1**) → the six-agent LangGraph pipeline → the FastAPI contract (**seam #2**) → operations console, citizen channels and officer artefacts. Every box decouples through the two seams; the spatial unit throughout is Uber H3 res-8 (~1 km hexagons).

The two seams

The entire system decouples through exactly two contracts, which is why three engineers could build in parallel for weeks without blocking each other:

- Seam #1 — the Supabase schema.** Models and agents **write** rows; the API only **reads** rows. No component calls another component directly.
- Seam #2 — the API contract.** Every endpoint returns one envelope: `{success, data, error, meta}`. The console, citizen channels and officer artefacts call only the API.

The consequence is operational as well as organisational: any layer can be replaced (a better forecast model, a new citizen channel, an additional city) without touching the others, and the whole platform runs offline from bundled fixtures in `DEMO_MODE` for demo-proof reliability.

The six AI agents

The pipeline is a LangGraph state machine, executed per city and traced per node. Routing is conditional: a **spike gate** escalates to enforcement only when cells actually spike, so the system never manufactures enforcement action on clean air.

Graph: orchestrator → attribution → forecast → **spike gate** → [enforcement, if spiking] → advisory → end, with a multi-city comparator running across cities.

Agent	Responsibility
A0 · Orchestrator	Reads the latest signals, identifies spiking cells, routes the graph, and stamps per-node latency traces that produce the signal-to-action metric.
A1 · Attribution	Per-cell source attribution — gradient boosting blended with chemical-signature priors, explained with SHAP, gated by an out-of-sample R^2 skill threshold below which it abstains.
A2 · Forecast	24/48/72-hour per-cell PM2.5 with conformalised prediction intervals and a persistence baseline retained for honest comparison.
Spike gate	Anomaly decision point on the forecast — escalates to enforcement only when warranted; otherwise routes straight to advisory.
A3 · Enforcement	Scores every registered and detected source (contribution × population exposed × actionability × model confidence), retrieves regulatory citations by RAG, and writes the ranked worklist with evidence.
A4 · Advisory	Deterministic, template-generated multilingual health advisories, escalated by ward vulnerability and audience segment (general, outdoor worker, elderly, school, respiratory).
A5 · Multi-City	Cross-city comparative intelligence — trends, dominant sources, compliance metrics and transferable intervention playbooks.

Supporting models: E1 satellite source detection · E2 dense 1 km concentration field (sparse stations → ~900 cells/city) · E3 counterfactual what-if engine · E5 inspector-hour optimiser · E7 health/₹/CO₂e impact layer · GRAP forecast-trigger · multimodal RAG over the NCAP/GRAP/CPCB regulatory corpus (text + satellite imagery).

SECTION 6

Functionality — the complete feature catalogue

6.1 Landing page

Public front door with live data: real-time Delhi AQI and the last measured pipeline latency fetched from the production API on load; a four-step explanation of the loop; the architecture diagram; twelve platform capability cards; a live production snapshot (source-mix breakdown, per-city current vs +24 h forecast, and the scale figures); a validation table stating each claim, its result and how it was checked; the full data-source list; and QR codes to open the app on a phone or subscribe to the Telegram bot.

6.2 Operations console — shell

- **City switcher** — Delhi, Bengaluru, Mumbai. Every layer (map, forecasts, worklists, advisories) follows the selection, which persists across visits.
- **Live status strip** — worst-cell AQI, category, PM2.5 and data freshness, with a live-connection indicator (WebSocket).
- **Compound-risk chips**, shown only when conditions are genuinely met: **Heat×Smog** watch/alert (IMD heatwave criteria × CPCB bands), **GRAP Stage I–IV from forecast** (our 24 h forecast enters a statutory CAQM band a day before observed AQI would trigger it), and **Dust×Traffic** co-occurrence (cells where both exceed 25% of blame — traffic resuspends construction dust).
- **Pipeline latency card** — wall-clock of the last full multi-agent run with a “signal→action < 5 min” badge.
- **Guided tour** — a four-step spotlight walkthrough on first visit, replayable at any time; seven-section navigation as a desktop sidebar and a thumb-reachable mobile tab bar; fully responsive from 320 px phones to ultrawide monitors; installable as a PWA.

6.3 Interactive blame map

- **Three view modes.** *Sources* — each hexagon coloured by its dominant source with opacity by dominance. *Satellite NO₂* — Sentinel-5P tropospheric column as independent evidence. *PM_{2.5}* — the dense 1 km modelled field, with a toggle between station-only and downscaled coverage and the measured skill note.
- **Four independent overlays.** Detected emission sources; wind plumes (Gaussian footprints oriented by the live wind field); administrative ward boundaries; and diesel-freight corridors with entry-window policy notes.
- **Cell Story** — clicking any hexagon opens the full per-cell narrative: the real place name (resolved offline from ward boundaries), source-share bars with confidence, the model's SHAP drivers in $\mu\text{g}/\text{m}^3$ with its own out-of-sample R^2 (or an explicitly labelled chemical-signature fallback), the 24/48/72-hour outlook with 80% intervals, and a one-click jump to the enforcement actions for that cell.

6.4 Enforcement intelligence (the officer loop)

- **Ranked workload** with category filters (construction, industrial, waste) and free-text search. When a cell is selected, the list re-sorts nearest-first; duplicate detections in one cell collapse into a single card so one inspection covers the cluster.
- **Each recommendation** shows its priority score, an officer-rubric score out of 10, the source's percentage contribution, residents exposed, the specific inspection to perform, and the regulatory basis.
- **Evidence dossier** — expands in place with a **real Sentinel-2 satellite patch** of the actual site (with detection confidence) plus up to three de-duplicated regulatory citations retrieved by RAG from the NCAP/GRAP/CPCB/Air Act corpus, each with a match score.
- **Draft Notice PDF** — one click generates an officer-ready notice: addressee block, findings, exposure assessment, cited statutory provisions, a 24-hour IST deadline, the embedded satellite image, a **projected-impact chart** (forecast PM_{2.5} with and without this source's contribution at +24/48/72 h, labelled a screening estimate), and an authorisation block. Every notice is stamped "**DRAFT — pending officer authorisation**"; nothing is ever auto-sent.
- **Status workflow and impact tracking** — proposed → approved → dispatched (or dismissed). Dispatching freezes the cell's 7-day PM_{2.5} baseline and opens a before/after measurement window, reporting the effect as the cell's change minus city-wide drift, marked provisional for the first week.
- **City intelligence panel** — registry source counts, traffic index, sensitive-zone counts and the mapped hospitals/schools driving advisory escalation.

6.5 Hyperlocal forecasting

- Horizon tabs for +24/+48/+72 h, charting the city forecast with its shaded 80% uncertainty band against the dashed persistence baseline.
- Backtested skill note (walk-forward, three folds) stating measured improvement over both persistence and climatology.
- Spike alerts for cells forecast at or above 90 $\mu\text{g}/\text{m}^3$, with a worst-first per-cell list.
- City statistics: 48-hour hourly PM_{2.5} trend from real station readings, live source-mix breakdown, and an AQI-band distribution showing what share of the city's cells sit in each CPCB category.

6.6 Citizen health advisory system

- **Four languages** — English, Hindi, Kannada and Marathi in native script, per city configuration.
- **Four delivery channels**, each previewable in-product: the web app, a live two-way **Telegram bot**, real **IVR phone calls** (both outbound and inbound — callers dial in and press 1/2/3 for Delhi/Bengaluru/Mumbai to hear the current advisory), and high-contrast **public-display** boards.
- **Vulnerability targeting** — 2,624 zones scored from OpenStreetMap hospitals, clinics, schools, elder-care and outdoor-work sites weighted by real gridded population; advisories escalate where forecast air is bad *and* sensitive people are.
- **Clean-air zones** — the lowest-PM_{2.5} cells right now with map directions, explicitly labelled as a modelled guide rather than a measurement.

- **Live broadcast** — an officer-confirmed action that sends a real Telegram message and places a real phone call, rate-limited server-side.
- Health text is **deliberately generated from deterministic templates, not an LLM**, so medical guidance cannot be hallucinated.

6.7 Multi-city comparative intelligence

Side-by-side comparison of all three cities: current versus +24 h averages, trend classification, dominant source, emission-signature match, annual health burden, transferable playbook recommendations, and enforcement compliance counts (approved / dispatched / dismissed) so administrators can see which interventions actually moved outcomes in comparable cities.

6.8 What-if simulator and inspector-hour optimiser

- Choose an intervention — waste-burn ban, halt construction dust, odd-even traffic restriction, industrial shutdown, or a combined GRAP Stage III package — and a horizon, then run the counterfactual over attribution × forecast.
- Returns average and best-cell ΔAQI, cells affected, confidence, and cited impact cards: **people protected** (from real gridded population), **health cost avoided (₹)**, **deaths averted** and **CO₂e co-benefit** — each figure carrying its citation.
- Intervention magnitudes are literature-grounded (Delhi odd-even trials ≈ 4–7% ambient PM2.5; CAQM GRAP stage schedules), not invented sliders.
- **Optimiser** — set an inspector-hour budget and receive ranked intervention packages by impact per inspector-hour.

6.9 Health and economic impact (City ROI) with fairness audit

- **City ROI** — annual attributable deaths, annual health burden in ₹, and what a 30% NCAP-target reduction would avert, with a burden-versus-avertable chart, a narrative funding case, and full citations. The rupee figure is explicitly labelled order-of-magnitude with its valuation caveat shown.
- **Fairness audit** — measured on every live recommendation: enforcement priority correlates with pollution contribution (the dominant driver, by design) and with population exposed (deliberate and disclosed). **No socio-economic input exists anywhere in the pipeline or schema** — the scorer sees only contribution, exposure, actionability and model confidence.

6.10 Live agent pipeline trace

A visual timeline of the last multi-agent run showing every node, its cumulative and per-step duration, and the spike-gate decision in plain language. When the air is clean, the enforcement node is shown as explicitly *skipped* rather than pretending to work. A **“Run agents live”** button triggers a real end-to-end execution and updates the trace and latency in front of the viewer.

SECTION 7

Data sources and technology

Data source	Used for
CPCB / CAAQMS (via data.gov.in, OpenAQ)	Real-time ground-truth pollutant measurements per station, mapped to H3 cells
Sentinel-5P (Copernicus)	Tropospheric NO ₂ column — independent satellite evidence of combustion activity
Sentinel-2 (Copernicus)	True-colour site imagery embedded as evidence in every enforcement dossier
NASA FIRMS	Thermal anomalies for biomass and waste-burning detection
Open-Meteo / ERA5	Wind, temperature and boundary-layer height driving forecasts and dispersion

OpenStreetMap (Overpass)	Emission-source registry, road network, freight corridors, and the facilities behind vulnerability scoring
GPW v4.11 (CIESIN/SEDAC, NASA)	Real population per cell — “people protected” and exposure are counts, not heuristics
GTFS + road network	Time-of-day mobility proxy feeding the attribution model

All sources are free and open. Ingestion runs on scheduled GitHub Actions (hourly measurements; daily model, registry and advisory refresh), keyless wherever possible.

Technology stack

Backend	Python · FastAPI on Render · LangGraph multi-agent orchestration · single response envelope
Frontend	React + TypeScript + Vite on Vercel · MapLibre GL + Deck.gl geospatial rendering · Recharts · Tailwind · installable PWA
Data platform	Supabase Postgres with PostGIS (geospatial) and pgvector (RAG embeddings), secured by row-level security
ML	Gradient boosting (LightGBM) with SHAP explanations · quantile regression with conformal calibration · Gaussian plume dispersion · super-resolution coverage downscaling · computer vision on satellite tiles
AI / retrieval	Multimodal RAG (text + imagery) over the NCAP/GRAP/CPCB corpus with vector search; model-agnostic LLM provider configuration
Channels	Telegram Bot API (two-way subscriptions) · Twilio Voice for inbound and outbound IVR · public-display rendering
Operations	GitHub Actions scheduled ingestion and refresh · CI on every push · offline DEMO_MODE from 17 bundled fixtures

API surface — 31 HTTP routes + 1 WebSocket

Group	Endpoints
Data feeds	/cities /aqi/current /history /attribution /forecast /coverage /plume /static-layers /mobility
Enforcement	/enforcement /enforcement/{id}/dossier /enforcement/{id}/notice.pdf POST /enforcement/{id}/status /interventions
Citizen	/advisory POST /advisory/broadcast /clean-zones /alerts/compound POST /telegram/webhook /ivr/inbound /ivr/advisory
Analysis	/comparison /roi POST /simulate POST /optimize /latency /traces
Agents / admin	POST /agent/query (runs the full graph) · POST /admin/cities (onboard a city, key-gated) · WS /live

SECTION 8

Validation — measured against the evaluation focus

Every claim below is reproducible from the validation notebook in the repository, including the failures we chose to keep visible.

Source attribution accuracy versus ground-truth emission inventories

Cosine similarity of our live attribution against published city inventories: **0.92 vs SAFAR-Delhi (2018), 0.88 vs CSTEP-Bengaluru (2022), 0.79 vs NEERI/Urban-Emissions Mumbai**, on locally attributable categories. Remaining discrepancies are explained rather than hidden — for example, live biomass burning is near zero in July, which is seasonally correct, whereas published inventories are annual averages. Attribution was additionally validated physically: traffic SHAP contribution is **2.30× higher during IST rush hours** with weather controlled.

AQI forecast accuracy at hyperlocal resolution (versus persistence)

Walk-forward backtests on live data, three folds. Skill = $1 - \text{RMSE}_{\text{model}} / \text{RMSE}_{\text{baseline}}$; both baselines are shown, weak results included.

City	Skill vs persistence (24 / 48 / 72 h)	Skill vs climatology (24 / 48 / 72 h)
Delhi (n ≈ 27k)	+4% / +4% / +8%	+31% / +18% / +16%
Bengaluru	+15% / +17% / +9%	+14% / +5% / -7%
Mumbai	+15% / +18% / +30%	+3% / +3% / +4%

Uncertainty was audited and corrected. Raw prediction intervals were found to under-cover (48–63% actual against a nominal 80%); conformalised quantile regression brought coverage to **75–80%**. A deep-learning forecaster (Temporal Fusion Transformer) was trained on GPU and **rejected** because gradient boosting won on held-out skill in all three cities — the honest result, kept rather than buried.

Enforcement recommendation quality rated by domain experts

Recommendations are scored against a CPCB/GRAP-derived six-dimension rubric, which the system’s own proxy rates **7–8/10 “would act”**. A structured independent review form (actionability, evidence quality, regulatory grounding, prioritisation, plus advisory relevance and an overall verdict) ships in the repository so a qualified reviewer — a pollution-control officer, environmental engineer or air-quality researcher — can replace that proxy with real expert judgement.

Citizen advisory relevance and language coverage

Advisories are generated for every ward from the forecast, tiered by CPCB breakpoints, escalated by measured vulnerability, and segmented by audience. Coverage is **four languages in native script** across **four delivery channels**, including inbound and outbound IVR for citizens who will never open a dashboard.

Demonstrated reduction in response time from signal to intervention

End-to-end latency is instrumented in production, not estimated: each agent node stamps the trace, and the measured signal-to-cited-action time is **0.8–9.7 seconds**, displayed live in the console and reproducible on demand via the “Run agents live” control. The status quo this replaces is measured in days.

Engineering quality

169 automated backend tests and 6 end-to-end browser tests, run in CI on every push, plus a fully offline demo mode that exercises the entire flow from bundled fixtures with zero live dependencies.

SECTION 9

Responsible AI — honest by construction

The platform is designed so that it cannot quietly overstate what it knows. These are engineering constraints, not aspirations:

- **The model abstains.** No ML-derived blame is published without out-of-sample skill (an R^2 gate); cells lacking observed markers shrink to the city mean rather than inventing an attribution.
- **Uncertainty is shown, not hidden.** Every forecast carries a calibrated interval and its persistence baseline; the coverage audit and its correction are documented.
- **Health advice cannot be hallucinated.** Citizen health text is generated from deterministic templates, never a language model.
- **Enforcement never over-claims.** Sources contributing under 2% never enter the worklist; citations are category-level and retrieved, not generated; notices are drafts requiring officer authorisation and are never auto-sent.
- **Nothing fabricated reaches production.** Impact figures return null rather than fall back on invented constants; satellite evidence is real imagery of the real site, and any generated visual is explicitly labelled as not being satellite imagery.
- **No socio-economic features exist** in the pipeline or the schema, so enforcement priority cannot encode income or land value; the fairness audit publishes exactly what does drive priority.
- **Every recommendation is auditable** after the fact through per-node agent traces, enforcement status history, and before/after intervention measurement corrected for city-wide drift.

SECTION 10

Business impact

Cited dose–response relationships and emission factors convert every modelled $\Delta\text{PM}_{2.5}$ into money, lives and carbon, so an administrator can defend a decision in budget terms.

Illustrative single intervention (waste-burn ban, 24 h)	Modelled result
Health cost avoided	₹2.56 million
People protected (real gridded population)	28,400
CO ₂ e co-benefit	382 tonnes

Delhi annual funding case (City ROI)	Modelled result
Premature deaths attributable per year	73,395
Annual health burden	₹3.67 lakh crore
Avertable per year by a 30% NCAP-target reduction	₹0.88 lakh crore

Sources carried in-product with every figure: WHO HRAPIE short-term $\text{PM}_{2.5}$ mortality functions as operationalised in WHO AirQ+; long-term concentration-response from Chen & Hoek (2020); India SRS crude death rate; UN World Urbanization Prospects population; IQAir annual concentrations; Andreae (2019) biomass emission factors. The rupee valuation is presented as order-of-magnitude with its caveat displayed, and CO₂e returns null where it cannot be defensibly computed.

Operational value beyond the headline numbers: inspector time is finite, and the optimiser ranks intervention packages by impact per inspector-hour; the forecast-triggered GRAP mapping fires a statutory graded response a day earlier than observed AQI would; and dispatched actions are measured afterwards, so a city accumulates evidence about which interventions actually work.

SECTION 11

Scalability, security and cost

- **A new city is one configuration file.** Adding a city means dropping a YAML file (bounding box, centre, languages, H3 resolution) into the city-config folder, or making one authenticated API call to the admin endpoint. Ingestion, models,

agents, API and UI are all city-agnostic — there is no per-city code anywhere in the system.

- **₹0 infrastructure today, with a clean paid path.** The entire platform runs on free tiers — Vercel, Render, Supabase and GitHub Actions. The same two-seam architecture scales horizontally onto paid tiers with no redesign.
- **Security by rows.** Row-level security is enabled across the schema: public data is readable through an anonymous JWT, every write path is service-role and server-side only, administrative operations sit behind a key header, and the voice webhooks are deliberately limited to public advisory text.
- **Resilience.** A complete offline mode serves the full experience from bundled fixtures, so a demo or a deployment never depends on an external service being awake.

SECTION 12

Running the project

The repository is offline-first: it runs end to end with no API keys.

<code>cp .env.example .env</code>	Keep <code>DEMO_MODE=true</code> to run entirely from bundled fixtures
<code>make install</code>	Python environment and web dependencies (CPU-only, lean)
<code>make dev</code>	Runs API and web together — API on :8000, app on :5173
<code>make test</code>	169 backend tests
<code>npx playwright test</code>	6 end-to-end browser tests (from the web directory)
<code>make prewarm</code>	Pre-demo smoke check across every endpoint, printing GO / NO-GO

Going live additionally requires linking a Supabase project and pushing the twelve migrations; full instructions, the API contract, the AI methodology, and a complete user guide covering every screen and control are included in the repository documentation.

SECTION 13

Roadmap

- **Physics upgrade** — integrate InMAP/PAVITRA source–receptor matrices beneath the what-if engine to move from linear-rollback screening to policy-grade emission-to-concentration modelling.
- **Evidence at scale** — complete training of the satellite computer-vision detector so source detection moves from labelled Earth-Engine heuristics to a full CNN pipeline.
- **Language reach** — in-language IVR voice, additional regional languages, and safely constrained LLM polish over the locked deterministic advisory facts.
- **Deeper enforcement integration** — municipal permit-registry connectors, automatic GRAP stage escalation into the worklist, and citizen-reported evidence with on-device vision.

SECTION 14

Team

Omkar Kadam · Sejal Kumbhar · Abhinav Prasad — Full-Stack AI Engineers.

Three engineers working across one shared codebase, each contributing across the entire stack — the machine-learning models, the multi-agent graph and API, and the application and citizen channels. The two-seam architecture is what made that possible: with the database schema and the API contract fixed as contracts, all three could build in parallel for weeks without any one blocking another. The architecture is also the team process.